

Instituto Superior de Agronomia, ULisboa

Master's in Green Data Science 2025-2026

Practical Machine Learning/Aprendizagem Automática Aplicada

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Final Project Guidelines

A. Project Proposal (Due June 12, 2026, Submit in Moodle)

Your project proposal should include the following information (even if tentative):

- **Problem Statement:** What problem will you be investigating? Why is it interesting?
- **Challenges:** What are the challenges of this project?
- **Dataset:** What dataset are you using? How do you plan to collect it? You can use your own data or gather data from online data repositories.
- **Method or Algorithm:** What method or algorithm are you proposing?
- **Evaluation:** How will you evaluate your results? What kind of analysis will you use to evaluate and/or compare your results (e.g., performance metrics or statistical tests)?

Step 1: Create a GitHub project repo, with `readme.md` file. Typically, it's a repo at the Github account of one of the group members.

In the project repo, create a PDF document or a markdown (MD) file called "project_proposal". The proposal should include the following:

- Project title
- Project category (e.g., tabular data, image classification, image segmentation, other—please specify)
- Full names and student IDs of team members (ideally two members)
- A 300–500-word description of your project plan

Step 2 (Individual submission): In Moodle, submit the link of the project proposal.

B. Project (Due June 26, 2026, Submit in Moodle)

Submission has 3 steps:

Step 1 of 3 (for the group, project repo)

Create your project repo: code and report.

Step 2 of 3 (individual, video)

Create an individual **short video** (that's no more than 5 minutes in length) in which you present your project. Your video should give an overview of the project aimed at the final user and should also highlight the main technical challenges from the developer's perspective – see grading guidelines.

Step 3 of 3 (individual, submit in Moodle)

In Moodle, submit the URL of the group's project repo and your individual video.

Report, code and data

1. **Report:** Your report should provide a comprehensive account of your project. It should be thorough yet concise, organized into the following sections:
 - **Introduction:** Motivation and explanation of the problem statement (you can reuse content from the project proposal).
 - **Data:** Description of the data, including any necessary cleaning and transformation steps. Identify data types and document data cleaning, feature selection, and feature engineering.
 - **Data Organization:** Description of training, validation, and test sets.
 - **Methods:** Description of the ML model(s) used, including hyperparameter and architecture choices.
 - **Results:** Presentation of results in tabular and/or graphical form.
 - **Analysis:** Analysis of results, including insights and discussions relevant to the project.
 - **Deployment** (optional): possibly as an app if it makes sense for your project
 - **References:** List of references used.
 - **Contributions:** A section detailing each team member's contributions to the project.

Format: If in the pdf format, this should be a ~4-6-page document, with additional pages for appendices and references if needed (the main document should be self-contained). Alternatively, this can be included in the same notebook as the code, but in that case be sure to include sections and subsections, so that a table of contents and links within the notebook are available.

2. **Code:** A Python notebook (with a Colab link) or script with the code. The code has to be available on your GitHub repository and clearly organized.

Use of AI: When submitting AI generated code, you must include comments of all prompts used to generate your code. Each comment must begin with the word '**prompt**' followed by the sections of code the prompt was used to generate. Include notes of any modifications you made to the generated code

3. **Data:** Include the dataset in the repository if it can be made available on GitHub, otherwise make the link available in the repository.

Grading (Up to 100 Points, After Discussion)

The project contributes 50% to the total grade. The final evaluation will rely on the following criteria that will be assessed through the video, report, and mostly from individual discussion with group members.

- **Video:** clarity (voice should be audible, with subtitles if necessary); video quality must be good. High: if the video is engaging and shows that you have a real understanding of the problem and the approach to solve it. Low: if the video is of low quality or if it only shows the app that was developed with no insights on the methods.
- **Novelty and Significance:** Importance and originality of the problem (e.g., a Kaggle problem may be significant but might lack novelty). High: address a problem significantly different from the ones discussed in class; Low: do a straightforward classification of a standard tabular data like the Iris data set for instance.
- **Report clarity:** Clear and concise presentation of the report.
- **Relevance:** Relevance of the project to the topics taught in class. High: the report refers to the relevant concepts discussed in class for the project at hand; Low: incomplete approach that misses relevant key aspects discussed in class.
- **Technical Quality:**

- **organization:** modularity, clear pipeline. High: clear pipeline using the appropriate classes from scikit learn or PyTorch; Low: no clear pipeline, making it difficult to understand if the code is correct.
 - **soundness:** use appropriate methods to address the problem and evaluate the quality of the solution
 - **validation:** follow correct procedures and address possible correlations among data observations
 - **Results and Conclusions:** Meaningfulness of the results and conclusions. High: engaging discussion of results; Low: show results with no overall discussion
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Examples of previous projects

- Identification of Greenhouses with Satellite Images (Image segmentation)
- Detecção de doenças em folhas de milho através de imagens (Image identification)
- Condicionantes socioambientais para as pirotregiones de Portugal continental (tabular data, clustering)
- Predicting covid-19 deaths in Portugal (tabular data, classification)
- App to help consumers to know more about the products they're considering to buy at a grocery store (image classification + database)
- BirdCLEF Competition (Kaggle). Identifying Eastern African Bird Species by Sound: develop machine learning models capable of accurately identifying bird species in Eastern Africa based on their sound recordings (sound data, classification)
- Predicting crop production from country, year, yield, crops, rainfall, temperature and pesticides with data from FAO and the World Data Bank (tabular data, regression)
- Identify grapevine varieties from images (image classification)
- App for bone fracture identification from X-ray images (image classification)
- Atmospheric Physics Climate Model, based on Kaggle competition "LEAP - Atmospheric Physics using AI" (tabular data, regression)
- PestTracker2: Identificação de praga de mosca-da-fruta (*Ceratitis capitata*) usando YOLOv8 (object detection on video)
- Estimation of soil salinity in rice production areas within mangroves from PlanetScope imagery with CNNs and RFs (image, regression)
- Identify from cellphone images the occurrence or not of trees in the foreground of the image (image classification)
- Creating a early fire detection model from ICNF fire occurrences (tabular data, classification)
- Genre classification of music tracks using the GTZAN dataset with using feature extraction and CNNs (sound data, classification)
- App for potato pest classification (image, classification)
- Air quality analysis from PM2.5 and PM10 concentration data (tabular data, classification)
- Identificação de pragas e doenças em tomateiros com recomendação de aplicações (image, tabular data, classification)
- Soil organic carbon predictor (tabular data, regression)
- Crop Recommendation (tabular data; multi-class classification)
- TreeID: Monitoring Portuguese Plant Diversity through a Collection-based Mobile Application: development on a mobile application that allows the user to collect vegetation species in augmented reality
- Previsão das propriedades de solos agrícolas através da utilização do sensor NIR (dados tabulares, regressão)